Scheduling Learned Image Codes over Fading Multi-User Downlinks: A Link Abstraction and a Calibration-Aware Restless-Bandit Scheduler

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***Abstract*—Deep joint source-channel coding (JSCC) is designed and evaluated on single links, yet a cellular downlink serves many users slot by slot, through a scheduler, before a deadline. Scheduling needs each slot's value to each user; no model provides it for a learned code. We show that a progressive learned code trained under slot-varying SNR obeys a link abstraction: its distortion is the source distortion plus a sensitivity-weighted sum of per-slot minimum mean-square-error terms. On two test sets it predicts held-out decodes within 0.055–0.083 dB, against 0.16–0.19 dB for EESM and 0.29–0.34 dB for MIESM given the same weights; a constant-SNR-trained code obeys no such rule. Calibrated from 12 decodes per image, the abstraction reduces each user to a two-number state, and deadline scheduling becomes a finite-horizon restless bandit. We propose CLASP, a calibration-aware, link-abstraction-based scheduler that solves it with exact per-user dynamic programs, a Lagrangian bound on every causal scheduler and an index policy, and rewards the probability of reaching a quality target under the measured error of its own prediction, from the abstraction and reports. Decoded, CLASP gains 0.5–0.6 dB of mean PSNR over proportional fair and brings an extra 8–15% of users to 30 dB over per-image rate selection, with fresh, noisy or aged reports. Against digital transmission with CQI-table link adaptation and HARQ it leads by 1.2–1.6 dB and 13–16% of users at 30 dB under 2 dB CQI errors or aged reports. Training the code on the channel CLASP creates adds 0.21 dB.**

***Index Terms*—Deep joint source-channel coding, hybrid automatic repeat request, image transmission, imperfect channel state information, index policies, Lagrangian relaxation, link abstraction, multi-user scheduling, restless bandits.**

I. Introduction

DEEP joint source-channel coding (JSCC) can map an image directly to channel symbols with a single learned encoder and decoder, eliminating the separate source and channel codes that digital coding needs. Trained end to end through the channel, deep JSCC beats digital transmission at low signal-to-noise ratio (SNR) and degrades gracefully as the channel worsens, instead of failing once the SNR drops below what a digital code was designed for [1], [2]. That makes it a strong candidate for image delivery in 6G, yet a learned code is designed and evaluated on a single link at a given SNR. A cellular downlink is a multi-user, multi-slot system with fading, imperfect channel knowledge and deadlines, so the gain a learned code reports on a single link [1], [3], [4] is not automatically the gain a base station delivers to many users over fading slots. A base station splits a frame of slots among many users through a scheduler, and each user's slots fade independently. Moreover, the base station knows only an estimate of the current slot's channel, making a channel-quality report [5] often late or wrong. Since each delivery has a deadline, a low-latency scheduler is necessary as well.

Existing approaches to serving images to many users over a fading downlink fall into two categories, namely *physical link design* and *the JSCC way*. The scheduler of the former serves the bits of a separate image codec, whereas that of the latter serves a learned code.

Physical link design is how cellular systems deliver images today. As illustrated in Fig. 1(a), its scheduler selects both the served users and the rate of their blocks in every slot. Link adaptation sets each rate from the channel quality indicator (CQI), hybrid automatic repeat request (HARQ) retransmits failed blocks, and proportional fair (PF) scheduling serves each user near the peak of its channel quality [6]–[8]. These designs assume that each block is either decoded or lost and produce the cliff effect, in which a slot below the design SNR of its block delivers nothing and any SNR above it adds no quality [1], [2]. Moreover, link adaptation and HARQ protect the bits from imperfect channel knowledge only by spending extra slots.

In contrast, JSCC design constructs the downlink on a learned code, whose distortion varies continuously with the SNR of its symbols. Hence no transmission is lost outright, and no coding rate has to be matched to an uncertain report. On a single link, per-image rate selection sends each image at the least rate at which its predicted quality meets a target [9]. As illustrated in Fig. 1(b), multi-user schedulers in this category value each user by such a predicted quality rather than by bits. They allocate the shared channel by the quality predicted at each user's SNR [10]–[12], train an allocation policy by deep reinforcement learning [13], [14], or embed such learning in a restless bandit over fading slots [15] (Section II-B).

Although these designs avoid the cliff effect of physical link design, they leave three challenges unresolved.

- The first challenge is the absence of a *link abstraction* for learned codes. The quality models of these designs, whether fitted curves or learned predictors [9], [10], [12], take a single SNR as input. The symbols of a scheduled image are instead spread over slots of different SNRs, and no rule is known for how these SNRs combine into the quality of a nonlinear code. Physical link design provides such rules for the error rate of a digital block, namely the exponential and mutual-information effective SNR mappings (EESM and MIESM) [16], [17]. To our

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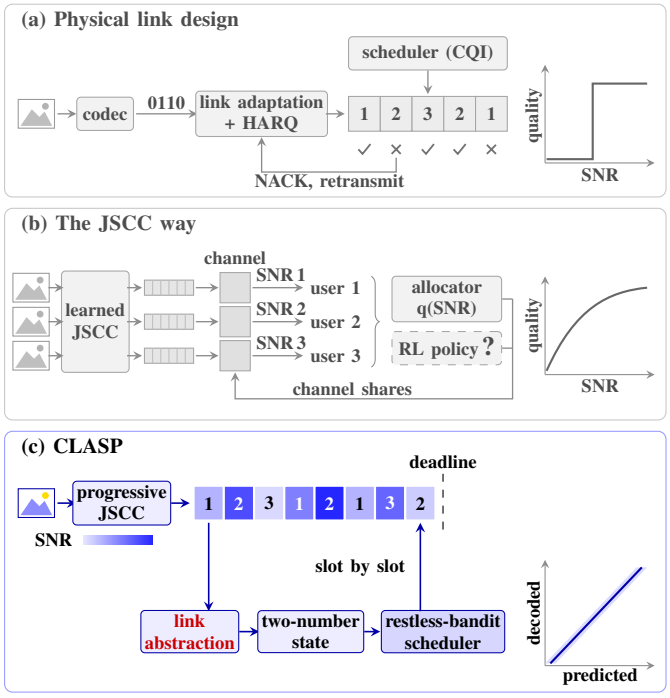


Fig. 1. Approaches to serving images to many users over a fading downlink. (a) Physical link design transmits the bits of a separate codec at a rate set from a channel report, so every block either decodes or fails. (b) The JSCC way transmits a learned code, and its scheduler values each user at a single SNR or follows a policy learned by reinforcement learning. (c) CLASP schedules a progressive code trained under slot-varying SNR, whose link abstraction values every slot. This restless-bandit scheduler serves every user slot by slot up to the deadline.

knowledge, neither mapping has been validated on the distortion of a learned code.

- The second challenge is *tractable and certifiable scheduling*. Choosing the users of every slot up to the deadline, without knowledge of future slots, is a stochastic dynamic program whose state grows exponentially with the number of users. The quality-based allocations are myopic, since they ignore the slots that remain in the frame [10]–[12]. Learned policies [13]–[15] require many training interactions and offer no certificate of their distance from the optimum.
- The third challenge is *imperfect information*, because the scheduler decides on predictions of both the quality and the channel. Per-image rate selection [9] treats its predicted quality as exact, so a user predicted just above a quality target may decode just below it. The learned restless-bandit scheduler likewise assumes exact channel knowledge [15]. Moreover, the advantage that motivates these designs was measured against an idealized digital link with exact channel knowledge and capacity-achieving codes [1], [18]. Whether this advantage persists under the reports of a real base station remains open.

All three challenges raise a fundamental question, namely whether a learned code admits a link abstraction. If such an abstraction exists, a further question is what makes a given code obey it.

To answer these questions, we trained a progressive code, which decodes from any prefix of its phases, under slot-varying SNR, and tested candidate rules on held-out decodes with mixed slot SNRs (Sections IV-B and V-A). We found three things. First, its distortion adds up over slots: the source distortion plus a sensitivity-weighted sum of per-slot minimum mean-square-error (MMSE) terms predicts held-out decodes with a mean absolute error of 0.055–0.083 dB on two test sets, against 0.16–0.19 dB for EESM and 0.29–0.34 dB for MIESM with the same per-phase weights. Linear analog codes obey this form by construction [19]; here a nonlinear learned code acquires it. Second, training decides it: the same architecture trained at one constant SNR per image obeys no such rule (the MMSE rule errs by 0.22 dB, and no rule does better than 0.15 dB), because it codes its later phases against the base reconstruction it expects at one SNR, the mismatch DeepJSCC-l observed for layers [20]. Third, the abstraction is cheap and accurate, yet its small error matters. Twelve decodes per image calibrate it, and it predicts the decoded quality of complete scheduled deliveries with a mean absolute error of 0.06–0.09 dB and a bias within ± 0.03 dB, with exact, noisy or aged reports. At a hard quality target, however, a scheduler that takes the calibrated curves as exact leaves an extra 1.7–2.0% of the users below 30 dB, at every cell size from 16 to 128 users. In short, training gives a learned code a link abstraction. With it, each user's state reduces to two numbers, the phases received and their accumulated weighted noise, and scheduling becomes a planning problem, provided the abstraction's residual error is part of the objective.

To address these challenges, we propose CLASP (Calibration-aware, Link-Abstraction-based Scheduling of Progressive codes), a scheduler for learned progressive codes in a multi-user fading downlink. Unlike existing JSCC methods, which value each user at one SNR or learn the allocation by reinforcement learning, CLASP values every slot for every user through the code's link abstraction and plans each user's delivery to the deadline exactly. It has one design for each challenge.

- *Link abstraction.* The progressive code is trained under slot-varying SNR and its MMSE link abstraction is calibrated per image, so that each user becomes a two-number state.
- *Scheduler.* On this state the frame becomes a finite-horizon restless bandit [21], a model already used to schedule layered video [22] and learned semantic codes [15]. A Lagrangian relaxation splits it into one small dynamic program per user under per-slot prices, solved exactly on a CPU; the relaxation's value is an upper bound on every causal scheduler; and users are served in the order of the Lagrangian index those programs define [23], [24].
- *Imperfect information.* Each program rewards the probability of reaching the quality target under a calibrated predictive distribution of the user's decoded quality, rather than a hard threshold on the predicted quality. The distribution's spread is the measured error of the prediction, from the calibrated curves and from the channel reports.

Given an aged report, each program decides on the posterior of the channel, not on the report itself.

The main contributions of this paper are as follows.

- *Discovery.* We find that a learned progressive code trained under slot-varying SNR obeys an additive MMSE link abstraction, more accurate than EESM and MIESM, while a code trained at constant SNR obeys none, and that a transmitter can calibrate it from 12 decodes per image. To our knowledge, effective-SNR mappings had not been tested on a learned code before.
- *Method.* We propose CLASP, which casts multi-user deadline scheduling of learned codes as a finite-horizon restless bandit with a two-number user state, solves it with exact per-user programs, a Lagrangian bound and an index policy, and is robust to the errors of its own quality prediction and of the channel reports.
- *Performance.* Decoded by the codec, CLASP gains 0.5–0.6 dB of mean PSNR over PF and brings an extra 8–15% of the users to 30 dB over per-image rate selection, with fresh, noisy or aged reports. In simulation its gap to the bound vanishes as the cell grows: from 0.21 to 0.005 dB per user between 8 and 128 users for mean PSNR, and for the target within $\pm 0.3\%$ of the bound from 16 users on. Against a digital chain with CQI-table link adaptation and HARQ, it leads at every CSI condition tested, by 1.2–1.6 dB and an extra 13–16% of the users at 30 dB with 2 dB CQI errors or reports aged to correlation 0.8. An idealized digital code 3 dB from capacity with the same HARQ leads CLASP by 2.4% of the users with exact reports and trails it by 5–14% from 1 dB CQI errors or correlation 0.95 on. Training the code on the channel CLASP creates adds 0.21 dB, decoded.

The rest of the paper is organized as follows. Section II reviews related work, and Section III states the system model and the problem. Section IV presents the training of the progressive code, its MMSE link abstraction, CLASP, and the training of the code on the channel CLASP creates. Section V evaluates the abstraction, and CLASP against PF, per-image rate selection, the Lagrangian bound and the digital chain. Section VI concludes and states the limitations.

II. Related Work

We review the two categories of Section I in turn, and then the scheduling theory CLASP builds on.

A. Physical Link Design for Multi-User Media Delivery

In a cellular downlink, an image is compressed once by a standard codec such as High Efficiency Video Coding (HEVC) [25], and its bits are carried by a physical layer that adapts to the channel. Adaptive modulation selects the constellation and the transmit power from the channel state, and its bit error rate depends on the error and the delay of the channel estimate it adapts to [6]; in Long Term Evolution (LTE), the receiver reports the channel as a 4-bit CQI [5]. HARQ retransmits a failed block, and the receiver either combines the noisy copies [26] or accumulates incremental redundancy; the throughput of such protocols over a fading Gaussian collision channel has closed forms [7]. Across users, a scheduler exploits multiuser diversity by serving each user when its instantaneous channel quality is near its peak, as PF scheduling does [8].

To evaluate such systems without simulating the decoding of every block, system-level simulators predict a block's error rate from the signal-to-interference-plus-noise ratios (SINRs) of all the resources it occupies. An effective SINR mapping compresses them into one effective SNR, from which the block error rate (BLER) is read off the link-level curve of an additive white Gaussian noise channel. EESM was derived for orthogonal frequency-division multiplexing (OFDM) from the union–Chernoff bound, with one parameter tuned per modulation and coding scheme [16]; MIESM maps the SINRs through the mutual information per received symbol or per bit [17]; and the 5G New Radio abstraction of the ns-3 simulator calibrates EESM and obtains SINR–BLER tables with a link-level simulator [27]. Both mappings predict whether a block decodes; Section V-A tests them as predictors of a learned code's distortion.

Media-aware schedulers value data by the distortion it removes. RaDiO reduces the rate-distortion optimized streaming of a presentation to the error-cost optimized transmission of each data unit, solved as a Markov decision process over the unit's transmission opportunities with a Lagrange multiplier pricing the rate, and essentially achieves the operational distortion-rate function of the source at the capacity of the packet erasure channel [28]. In a wireless downlink with several users, gradient-based schedulers set each user's rate from its channel quality and the gradient of a utility built from the distortion of its received video [29]. Hosseini and Panwar model scalable video streaming to many users over Markov channels as a restless bandit, relax it to a linear program whose value they plot beside their policies, and improve network utilization by up to a factor of two over PF [22]. For packets that expire at the end of each frame, Hou, Borkar and Kumar characterize the feasible per-client delivery ratios and show that serving the client with the largest delivery debt first fulfills every feasible set [30]. All of these treat a block or a packet as decoded or lost, so the channel affects quality only through how many of them get through, the source of the cliff effect that learned codes avoid [1]. Our link abstraction takes the place of an effective SNR mapping for a learned code: it predicts distortion rather than a probability of decoding, and so it can value every slot a user receives.

B. The JSCC Way

Deep JSCC trains a convolutional encoder and decoder through a non-trainable channel layer. It outperforms JPEG or JPEG2000 compression with a capacity-achieving channel code at low SNR and small bandwidth, has no cliff effect, and degrades gracefully when the SNR departs from its training value [1]. Later codes adapt one model to the channel and the budget. Channel-wise attention scales the features by the SNR, so that one model serves a range of SNRs [3]; nonlinear transform source-channel coding learns an entropy model of its latent representation and adapts the transmission rate to

it [31]; Swin Transformer backbones adapt to the channel state and the target rate within a single model [4]; and predictive and adaptive deep coding (PADC) predicts each image's PSNR from its content, the SNR and the compression ratio, and sends it at the least compression ratio that meets a quality constraint [9]. Other codes exploit the structure of the channel: OFDM over multipath fading [32], multiple-input multiple-output (MIMO) channels, with robustness to channel estimation errors [33], and channel output feedback, which refines the receiver's estimate block by block and shortens variable-length transmission [18], [34]. DeepJSCC-l sends an image in layers, as successive refinements or as multiple descriptions, with negligible loss against a single transmission [20]. A tutorial on semantic and task-oriented communications is given in [2].

These codes are compared with separation-based transmission on single links: against JPEG or JPEG2000 with a capacity-achieving code [1], against Better Portable Graphics (BPG) with low-density parity-check (LDPC) codes over OFDM, with 2.5–4 dB of SNR gain at equal quality [32], and against BPG with 5G LDPC codes [4]. In some of these comparisons the digital side is idealized, with capacity-achieving codes and exact channel knowledge at the receiver [1], [18]; Section V-D compares instead with CQI-driven link adaptation and HARQ under erroneous and aged reports.

Linear analog transmission is the precedent for our link abstraction. SoftCast makes the network stack act as a linear transform of the pixels, so that channel noise reduces sharpness instead of causing glitches, and one multicast stream gives each receiver a quality commensurate with its channel [19]. For a linear code, the distortion is a sum of per-component MMSE terms by construction; the learned code of Section IV-B acquires this form, but only when it is trained for slot-varying SNR.

To serve many users, semantic and JSCC systems allocate the shared channel by each user's predicted quality: channel assignment and the number of transmitted semantic symbols from a semantic spectral efficiency [10]; user association and bandwidth from a message-level throughput [11]; compression ratios and transmission rates by Bayesian optimization over Gaussian-process models of quality against SNR and compression ratio [12]; and, in a non-orthogonal uplink shared with a bit-based user, the communication mode of the semantic user in each fading state [35]. Multi-user semantic transceivers are also trained end to end for image retrieval, machine translation and visual question answering [36]. Where the allocation changes over time, it is learned by deep reinforcement learning: proximal policy optimization with an attention network assigns resource blocks and the part of each user's semantic content to send [13], and a deep deterministic policy gradient agent sets each user's compression ratio, transmit power and bandwidth [14]. The allocations value each user's transmission at one SNR and decide it as a whole, so none of them can value a slot for an image whose symbols are spread over slots of different SNRs; the learned policies need many interactions to train and come with no bound on their distance from the optimum.

C. Restless Bandits and Lagrangian Index Policies

Whittle relaxed the constraint on the number of projects operated at each instant to a constraint on its expected value. The Lagrange multiplier of the relaxed constraint defines an index, and Whittle conjectured that operating the projects of largest index is nearly optimal when there are many of them [21]. Weber and Weiss proved the conjecture when the fluid approximation of the index policy has a globally stable equilibrium point, and gave an example in which it fails [37]. More generally, dynamic programs whose subproblems are coupled only through constraints on their actions admit Lagrangian and linear-programming relaxations whose value functions separate by subproblem [38]. Relaxing the limit on the number of items selected at a time yields both an upper bound on the optimal value and the Lagrangian index policy, and under mild conditions both are asymptotically optimal with many items, with tie-breaking playing an essential role [24]. A linear-program-based index policy is asymptotically optimal over finite and infinite horizons, at a square-root rate in general and exponentially fast under further conditions [39]. For deadline scheduling, Hao, Xu and Tong define the Lagrangian priority value of a job as the greatest tax under which it is optimal to process it, and prove that serving jobs in that order is asymptotically optimal as the system scales [23].

The same tools are established in wireless scheduling. For dynamic multichannel access, the Whittle index has a closed form, and for non-identical channels a Lagrangian relaxation gives an upper bound on performance [40]. For the age of information of clients behind unreliable channels, Whittle's index policy performs near optimally [41]. For remote inference, the joint choice of feature lengths and transmissions forms a restless multi-armed, multi-action bandit [42]. Closest to ours, McDowell et al. schedule a learned semantic code over independent Rayleigh-fading slots: a Lagrangian relaxation with one price splits the problem into per-sensor programs whose values are learned by reinforcement learning, the reward is a task value computed from the decoder's posterior, and a fluid upper bound is stated but not plotted [15]. Each slot, however, carries a fresh codeword of the newest observation, so nothing accumulates across slots toward a deadline, and the channel state is known exactly.

Restless bandits thus already schedule layered video [22] and learned semantic codes [15], which supports casting the deadline scheduling of learned codes as one. CLASP imports this machinery: the Lagrangian relaxation with one price per slot, its upper bound, and the Lagrangian priority policy [23], [24]. What is new is the arm and the reward: each user's phases accumulate over slots of different SNRs toward a decoded quality at the deadline, the link abstraction reduces each user to a two-number state whose program is solved exactly, and the reward accounts for the measured error of the quality prediction.

III. Preliminaries: System Model and Problem Statement

This section defines the downlink frame, the channel and the reports the base station receives, the progressive code,

TABLE I
NOTATION. VALUES ARE THE DEFAULTS OF SECTION V; $\mathcal{U}[a,b]$ IS THE UNIFORM DISTRIBUTION ON $[a,b]$.

Symbol	Meaning	Value
K, $\mathcal{K}$, k	number, set and index of users	$K = 8$
T, t	deadline in slots, slot index	$T = 24$
B	resource units per slot	1
B_t	units available in slot t	
p	probability that a unit is taken	0
S	complex channel uses per unit	$3HW/48$
$a_{k,t}$	1 if user k is served in slot t	
$\mathbf{a}_t$	schedule $(a_{1,t}, \dots, a_{K,t})$ of slot t	
$\bar{\gamma}_k$	mean SNR of user k	$\mathcal{U}[0, 15]$ dB
$h_{k,t}$	fading coefficient	$\mathcal{CN}(0,1)$
$\gamma_{k,t}$, $n_{k,t}$	slot SNR, noise variance	
$\gamma_{\min}$, $\gamma_{\max}$	SNR range of training	-2, 22 dB
$\hat{\gamma}_{k,t}$, $\hat{h}_{k,t}$	reported SNR, reported fading	
$\epsilon_{k,t}$, σ_ϵ	report error, standard deviation	σ_ϵ = 0–3 dB
r, f_D, τ	correlation, Doppler frequency, age	r = 1–0
$\mathbf{x}_k$, $\hat{\mathbf{x}}_k$	image, its reconstruction	
H, W	image height and width in pixels	
f, g	encoder, decoder	
P, j	phases per code, phase index	$P = 6$
$\mathbf{u}_{k,j}$, $\mathbf{z}_{k,j}$	phase j, as sent and as received	
$t_{k,j}$	slot that carries phase j	
L_k	phases held at the deadline	$L_k \le P$
$\gamma_{k,(j)}$, $n_{k,(j)}$	SNR, noise variance of phase j	
D_k, q_k	MSE, PSNR at the deadline	
$D_\emptyset$	MSE with no phase received	0.1
c, η_c	CQI index, its efficiency	[5]
θ_c	SNR threshold of index c	-6.7–22.7 dB
$R(\cdot)$	rate selected for an SNR	
$\eta_{k,t}$	rate of the block in slot t	
β	rate backoff	tuned
M	transmissions per block, HARQ	4
$I(\cdot)$	information per transmission	
Γ	SNR gap of the idealized code	3 dB
b_k	bits per pixel received	
π, $\mathcal{I}_t$	policy, information at slot t	
Q	quality target	30 dB
J_{mean}, J_Q	mean PSNR, share at the target	

the digital chain it is compared with, and the scheduling problem. Table I collects the notation, with the default values of Section V, and Fig. 2 illustrates the frame and the code.

A. Downlink Frame

A base station serves K users, indexed by $k \in \mathcal{K} = \{1, \dots, K\}$, over a frame of T slots, indexed by $t = 1, \dots, T$. Each user is owed one image $\mathbf{x}_k \in [0,1]^{3\times H\times W}$, of $H \times W$ pixels with three color channels and intensities scaled to $[0, 1]$, and every image must be delivered by the end of slot T, the deadline. Each slot holds B resource units of S complex channel uses each: $B = 1$ is time-division multiple access, and $B > 1$ models B groups of subbands within one slot. A user receives at most one unit per slot. Other traffic may take a unit: each unit is available to the frame with probability $1-p$, independently of the other units and slots, so the number of units available in slot t is

$$B_t \sim \text{Binomial}(B, 1-p), \tag{1}$$

and B_t becomes known at the start of slot t. The schedule is a set of binary variables $a_{k,t} \in \{0, 1\}$, with $a_{k,t} = 1$ if user k receives a unit in slot t. Since a user takes at most one unit, a slot can serve at most B_t users:

$$\sum_{k=1}^{K} a_{k,t} \le B_t, \quad t = 1, \dots, T. \tag{2}$$

B. Channel and Channel Reports

The channel of each user is flat within a slot and changes independently from slot to slot (block fading). In slot t, a complex symbol s of unit average power sent to user k arrives as

$$y = \sqrt{\bar{\gamma}_k}\, h_{k,t}\, s + w, \tag{3}$$

where $\bar{\gamma}_k$ is the mean SNR of user k, set by its path loss; $h_{k,t} \sim \mathcal{CN}(0,1)$ is its Rayleigh fading coefficient, independent over users and slots; and $w \sim \mathcal{CN}(0,1)$ is the receiver noise, independent over symbols. $\mathcal{CN}(\mu, v)$ denotes a circularly symmetric complex Gaussian law of mean μ and variance v. The receiver knows $h_{k,t}$ and equalizes, $z = y/(\sqrt{\bar{\gamma}_k}\, h_{k,t}) = s + w/(\sqrt{\bar{\gamma}_k}\, h_{k,t})$, so every symbol of the slot crosses an additive white Gaussian noise (AWGN) channel with SNR $\gamma_{k,t}$ and noise variance $n_{k,t}$:

$$\gamma_{k,t} = \bar{\gamma}_k\, |h_{k,t}|^2, \qquad n_{k,t} = \frac{1}{\gamma_{k,t}}. \tag{4}$$

As $|h_{k,t}|^2$ is exponential with unit mean, $\gamma_{k,t}$ is exponential with mean $\bar{\gamma}_k$. A superscript dB marks a value in decibels, as in $\gamma_{k,t}^{\text{dB}} = 10\log_{10}\gamma_{k,t} = \bar{\gamma}_k^{\text{dB}} + 10\log_{10}|h_{k,t}|^2$.

At the start of slot t, the base station holds a report $\hat{\gamma}_{k,t}$ of the slot SNR of every user, and nothing about later slots. Three report models are considered. *Exact reports* give $\hat{\gamma}_{k,t} = \gamma_{k,t}$. *Erroneous reports* are off by a Gaussian error in decibels,

$$\hat{\gamma}_{k,t}^{\text{dB}} = \gamma_{k,t}^{\text{dB}} + \epsilon_{k,t}, \qquad \epsilon_{k,t} \sim \mathcal{N}(0, \sigma_\epsilon^2), \tag{5}$$

independent over users and slots, where $\mathcal{N}(\mu, v)$ denotes a real Gaussian law of mean μ and variance v, so that σ_ϵ is the standard deviation of the error. *Aged reports* were measured a time τ before the slot, and the channel has changed since:

$$\hat{h}_{k,t} = r\, h_{k,t} + \sqrt{1-r^2}\, e_{k,t}, \qquad \hat{\gamma}_{k,t} = \bar{\gamma}_k\, |\hat{h}_{k,t}|^2, \tag{6}$$

where $\hat{h}_{k,t}$ is the fading coefficient the report was measured on, $e_{k,t} \sim \mathcal{CN}(0,1)$ is independent of $h_{k,t}$, and $r = |J_0(2\pi f_D \tau)| \in [0,1]$ is the magnitude of the correlation of the fading process at lag τ under Jakes' isotropic-scattering model [43], with f_D the Doppler frequency and J_0 the Bessel function of the first kind and order zero. The magnitude suffices: as $e_{k,t}$ is circularly symmetric, the joint law of $|h_{k,t}|$ and $|\hat{h}_{k,t}|$, which is all the SNRs depend on, is the same for a correlation of either sign. A report with $r = 1$ is fresh, and one with $r = 0$ says nothing about the slot. Since $\hat{h}_{k,t}$ has unit variance and $\mathbb{E}[h_{k,t}\hat{h}_{k,t}^*] = r$, where * denotes complex conjugation, the two coefficients are jointly Gaussian with correlation r, and the channel given the report is

$$h_{k,t} \mid \hat{h}_{k,t} \sim \mathcal{CN}\big(r\,\hat{h}_{k,t},\, 1-r^2\big), \tag{7}$$

which a transmitter that knows r can use instead of the report itself (Section IV-D). In the evaluation, slot SNRs and reports are clipped, in decibels, to the range $[\gamma_{\min}, \gamma_{\max}]$ on which the code was trained.

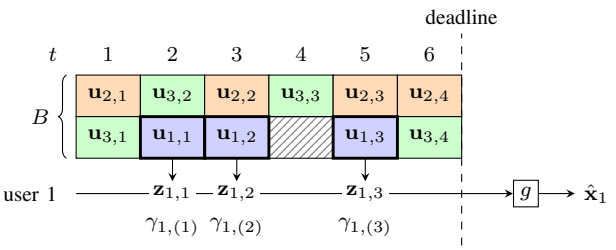


Fig. 2. A frame with $K = 3$ users, $B = 2$ units per slot and $T = 6$ slots. Each cell is a resource unit, shaded by the user it serves and labeled with the phase it carries; the column of slot t is the schedule $\mathbf{a}_t$, chosen at the start of the slot from $\mathcal{I}_t$. Other traffic takes the hatched unit, so $B_4 = 1$. User 1 is served in slots 2, 3 and 5, so $t_{1,1} = 2$, $t_{1,2} = 3$, $t_{1,3} = 5$ and $L_1 = 3$; its phases arrive as $\mathbf{z}_{1,j}$, each at the SNR $\gamma_{1,(j)} = \gamma_{1,t_{1,j}}$ of its slot, and the decoder g reconstructs $\hat{\mathbf{x}}_1$ from them at the deadline.

C. The Progressive Code

Each image is encoded once, before the frame, by an encoder f that maps it to P phases of S complex symbols each,

$$f(\mathbf{x}_k) = (\mathbf{u}_{k,1}, \dots, \mathbf{u}_{k,P}), \qquad \mathbf{u}_{k,j} \in \mathbb{C}^S, \tag{8}$$

normalized to unit average power per complex symbol. A phase fills one resource unit, so it has a bandwidth ratio, the number of complex channel uses per real source dimension, of $S/(3HW)$, and the whole code has $PS/(3HW)$; with the values of Table I, these are $1/48$ and $1/8$. The phases are sent in order: the j-th unit that user k receives carries $\mathbf{u}_{k,j}$. User k receives $L_k = \sum_{t=1}^{T} a_{k,t}$ units in the frame, at most P, so it holds L_k phases at the deadline, and for $j = 1, \dots, L_k$ the slot that carries phase j is

$$t_{k,j} = \min\Big\{t : \sum_{t'=1}^{t} a_{k,t'} = j\Big\}, \tag{9}$$

the first slot by which user k has been served j times (Fig. 2). Phase j arrives after equalization as

$$\mathbf{z}_{k,j} = \mathbf{u}_{k,j} + \mathbf{w}_{k,j}, \qquad \mathbf{w}_{k,j} \sim \mathcal{CN}\big(\mathbf{0},\, n_{k,(j)}\mathbf{I}_S\big), \tag{10}$$

where $\mathbf{I}_S$ is the $S \times S$ identity matrix, so the S entries of the noise $\mathbf{w}_{k,j}$ are independent with variance $n_{k,(j)} = n_{k,t_{k,j}}$, that of the slot that carried the phase. Each phase thus meets the SNR of its own slot, $\gamma_{k,(j)} = \gamma_{k,t_{k,j}}$. At the deadline, user k decodes the prefix it holds with one decoder g that accepts any number of phases,

$$\hat{\mathbf{x}}_k = g(\mathbf{z}_{k,1}, \dots, \mathbf{z}_{k,L_k}), \tag{11}$$

and neither f nor g is told an SNR. The quality of the reconstruction is its mean squared error (MSE) per pixel and color channel, D_k, or its peak signal-to-noise ratio (PSNR), q_k:

$$D_k = \frac{1}{3HW}\big\|\mathbf{x}_k - \hat{\mathbf{x}}_k\big\|_2^2, \qquad q_k = 10\log_{10}\frac{1}{D_k}\ \text{dB}, \tag{12}$$

where the peak intensity is 1. A user that receives no phase is assigned $D_k = D_\emptyset$, a PSNR of 10 dB for the value of Table I. The decoded quality thus depends on the image, on the number of phases received and on the SNR each of them met. For a trained, nonlinear decoder no closed form of this dependence is known; Section IV-B models it.

D. The Digital Baseline

Physical link design sends the image as bits. The image is compressed into HEIC files, which use HEVC intra coding [25], at several sizes. A unit given to user k in slot t carries one block of S channel uses at a spectral efficiency $\eta_{k,t}$, in bits per channel use, set from the report through the CQI table:

$$\begin{aligned} R(\gamma^{\mathrm{dB}}) &= \max\big\{\eta_c : \theta_c \le \gamma^{\mathrm{dB}}\big\}, \\ \eta_{k,t} &= R\big(\hat{\gamma}_{k,t}^{\mathrm{dB}} - \beta\big). \end{aligned} \tag{13}$$

Here η_c is the efficiency of CQI index $c = 1, \dots, 15$ in Table 7.2.3-1 of [5], from 0.15 to 5.55 bits per channel use. A CQI index is defined there as one whose transport block could be received with an error probability of at most 0.1, and θ_c is the AWGN SNR we take for that, from -6.7 dB for $c = 1$ to 22.7 dB for $c = 15$. Below θ_1, $R = 0$ and the unit carries nothing. The backoff β, in decibels, trades the rate of a block against the risk that a report error makes it fail: with exact reports $\beta = 0$, and otherwise Section V tunes it for each scheduler and objective. A block is decoded if the true SNR supports its rate, $R(\gamma_{k,t}^{\mathrm{dB}}) \ge \eta_{k,t}$, and is lost otherwise; an acknowledgment tells the base station which.

With hybrid automatic repeat request (HARQ), a lost block is sent again in each of the user's next slots until it is decoded or has been sent M times, when it is dropped; a block still pending at the deadline is lost. After m transmissions, at SNRs $\gamma_1, \dots, \gamma_m$, the receiver combines them. With incremental redundancy [7] the block is decoded once

$$\sum_{i=1}^{m} I(\gamma_i) \ge \eta, \tag{14}$$

and with Chase combining [26] once

$$I\Big(\sum_{i=1}^{m} \gamma_i\Big) \ge \eta, \tag{15}$$

where η is the rate of the block, set at its first transmission, and $I(\gamma)$, the information one transmission adds, is the efficiency of the CQI table interpolated linearly in decibels between its entries; below θ_1 it falls with the slope of the first segment, to zero at $\theta_1 - 3.7$ dB, and above θ_{15} it stays at η_{15}. Because $I(\gamma)$ equals η_c at θ_c and increases strictly up to θ_{15}, a single transmission is decoded exactly when $R(\gamma^{\mathrm{dB}}) \ge \eta$, as without HARQ.

As with the code, a user is served in at most P slots, so that both deliver an image in at most PS channel uses. At the deadline user k has received

$$b_k = \frac{S}{HW} \sum_{\text{decoded blocks}} \eta \tag{16}$$

bits per pixel, and it decodes the largest file of $\mathbf{x}_k$ whose size, in bits per pixel, is at most b_k, or nothing ($D_k = D_\emptyset$) if even the smallest file is larger. An idealized digital code replaces both R and I by $\log_2(1+\gamma/\Gamma)$, the rate of a code whose SNR gap to capacity is Γ, with the same backoff and HARQ.

E. Problem Statement

A scheduler is a policy $\pi = (\pi_1, \dots, \pi_T)$ that chooses the schedule of slot t from what the base station knows at its start:

$$\mathbf{a}_t = \pi_t(\mathcal{I}_t), \tag{17}$$

where $\mathbf{a}_t = (a_{1,t}, \dots, a_{K,t})$ is the schedule of slot t and

$$\mathcal{I}_t = \big\{B_{t'},\, \hat{\gamma}_{k,t'} : k \in \mathcal{K},\ t' \le t\big\} \cup \big\{\mathbf{a}_{t'} : t' < t\big\} \tag{18}$$

holds the units available and the reports up to slot t, and the schedules before it. From $\mathcal{I}_t$ the base station knows how many phases each user holds and in which slots they were sent, but not the SNRs they met. Besides $\mathcal{I}_t$, it knows the images and the mean SNRs $\bar{\gamma}_1, \dots, \bar{\gamma}_K$ from the start of the frame; a transmitter that accounts for aged reports also knows r; and the digital chain knows the acknowledgments of its past blocks. The policy is causal: it never sees the report of a later slot. Two objectives are considered. The first is the mean PSNR at the deadline,

$$J_{\text{mean}}(\pi) = \mathbb{E}\Big[\frac{1}{K}\sum_{k=1}^{K} q_k\Big], \tag{19}$$

and the second is the share of users whose image reaches a quality target Q by the deadline,

$$J_Q(\pi) = \mathbb{E}\Big[\frac{1}{K}\sum_{k=1}^{K} \mathbf{1}\{q_k \ge Q\}\Big], \tag{20}$$

where $\mathbf{1}\{\cdot\}$ is the indicator function. The expectations are over the fading, the reports, the available units and the channel noise, and in the evaluation also over the images and the mean SNRs drawn for each frame. The scheduling problem is

$$\max_{\pi \text{ causal}} \ J(\pi) \quad \text{s.t.} \quad (2), \quad a_{k,t} \in \{0,1\}, \quad L_k \le P \ \ \forall k, \tag{21}$$

with $J = J_{\text{mean}}$ or $J = J_Q$.

Problem (21) is a finite-horizon stochastic control problem with three difficulties. First, each user's reward is the decoded quality q_k of (12), for which a learned code offers no closed form. Second, the state at slot t holds every user's delivery history, the number of phases it holds and the SNR each met, so it is continuous, its dimension grows with K, and the users are coupled through (2). Third, the decisions are made on the reports $\hat{\gamma}_{k,t}$, not on the SNRs $\gamma_{k,t}$ that decide the quality. Section IV takes them in turn: a link abstraction predicts q_k and reduces each user's history to two numbers (Section IV-B); the problem then becomes a restless bandit [21], which CLASP relaxes into one exact dynamic program per user (Section IV-C); and a calibration-aware reward and the posterior (7) account for the reports (Section IV-D).

IV. Method

A. Training the Progressive Code for Slot-Varying SNR

B. The MMSE Link Abstraction

C. CLASP: Deadline Scheduling as a Finite-Horizon Restless Bandit

D. Imperfect Information: The Calibration-Aware Reward and Aged Reports

E. Training the Code on the Channel CLASP Creates

V. Experiments

A. Link Abstraction Accuracy

B. Mean PSNR: Near the Bound, and over PF

C. Quality Target and Latency

D. Imperfect CSI and the Digital Baseline

E. Training the Code on CLASP's Channel

F. Ablations and System Choice

VI. Conclusion

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